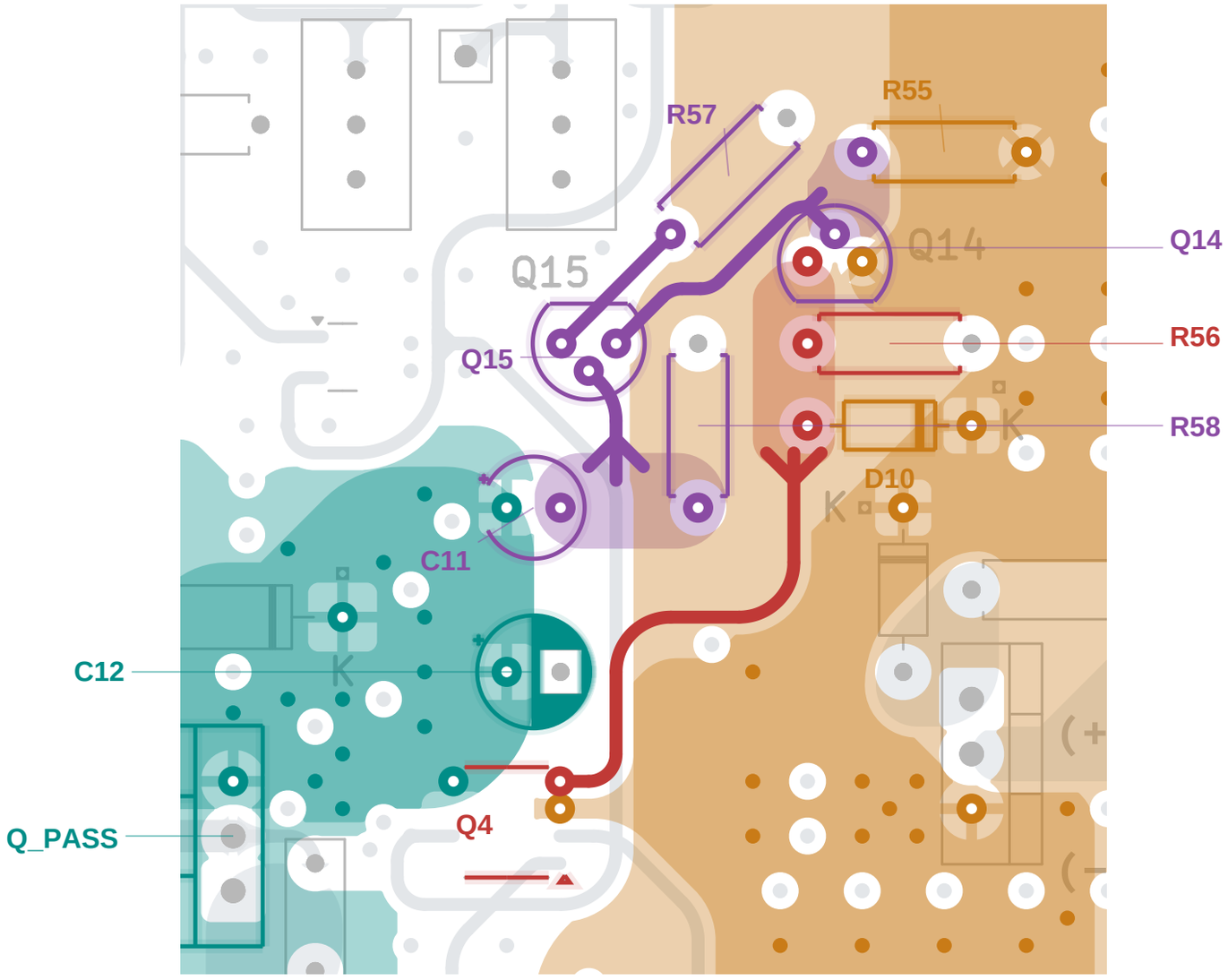
Startup ramp and series switch

Actual KiCad copper + component highlights | Top-side projection of both outer layers



Protected supply

VEE supplies the startup stage and Q4B pin 3.

Ramp feedback

C11 couples changes of VCTRL into the Q15 / Q14 control network.

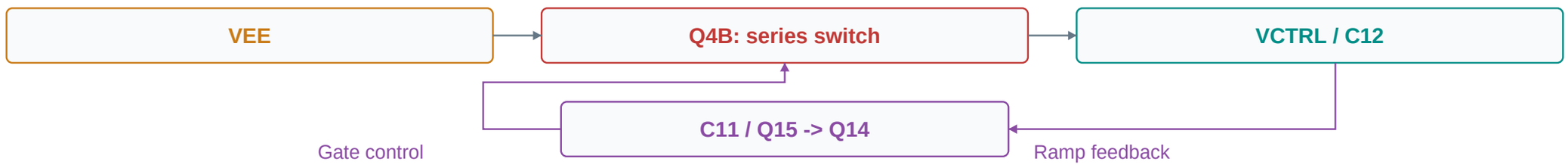
Switch gate

The Q14-controlled node drives Q4B gate pin 4; D10 connects it to VEE.

Ramped supply

Q4B pin 5 feeds VCTRL, C12 and the pass emitter. C12 is 10 uF in the PCB.

Q4B is the startup section of the shared Q4 package. The enable section Q4A is a separate path.



Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.